

Erdős Problem #770

EP-770

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $h(n)$ be minimal such that $2^n - 1, 3^n - 1, \dots, h(n)^n - 1$ are mutually coprime.

Does, for every prime p , the density δ_p of integers with $h(n) = p$ exist? Does $\liminf h(n) = \infty$? Is it true that if p is the greatest prime such that $p - 1 \mid n$ and $p > n^\epsilon$ then $h(n) = p$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>